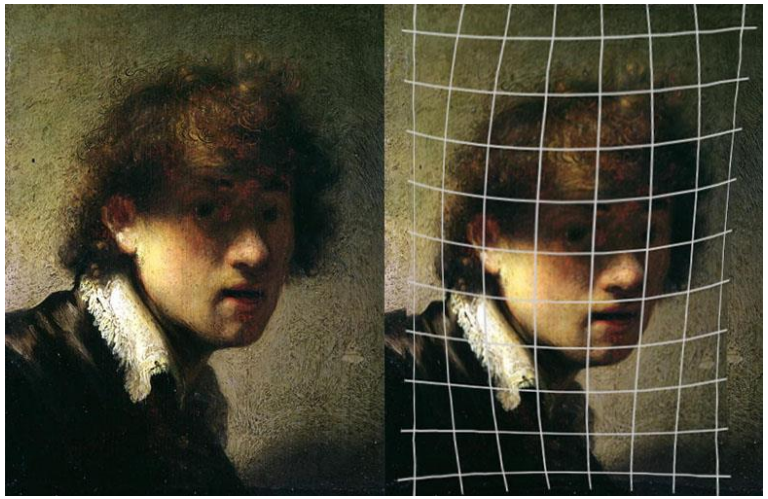


Nonlinear Warp Fields as a Source of Physiological Information in fMRI

05/22/26 Burak Akin

SFIM





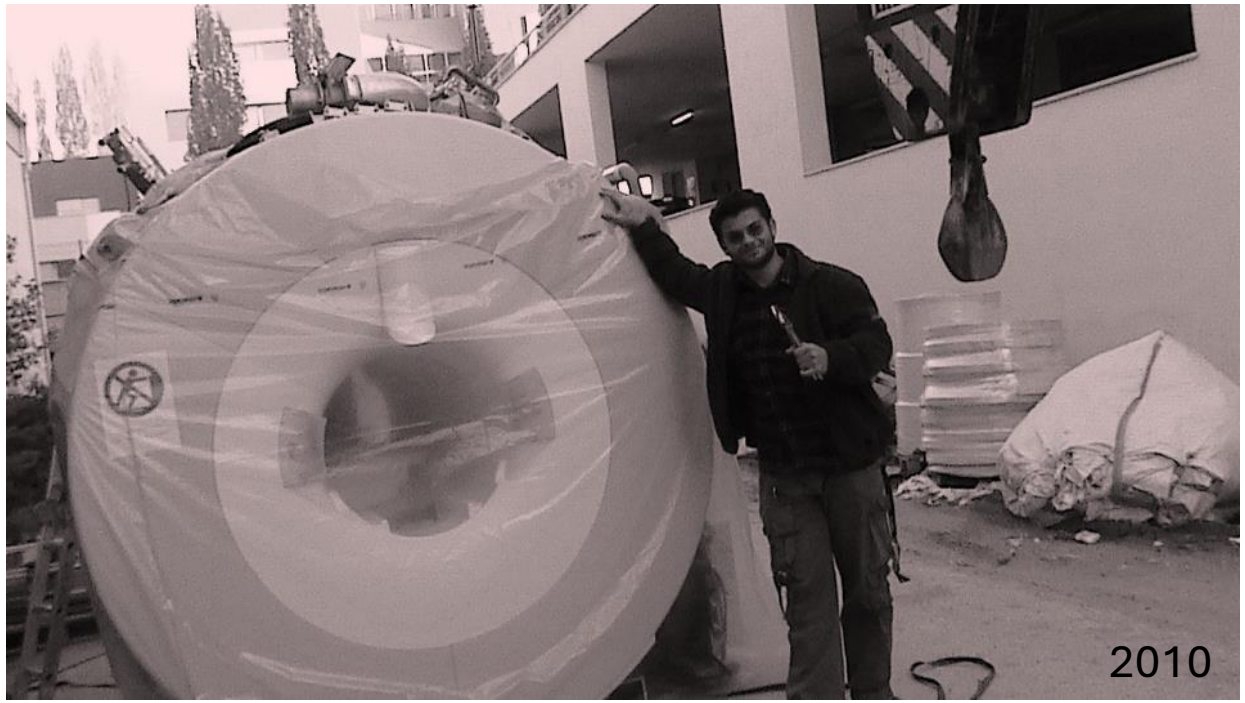


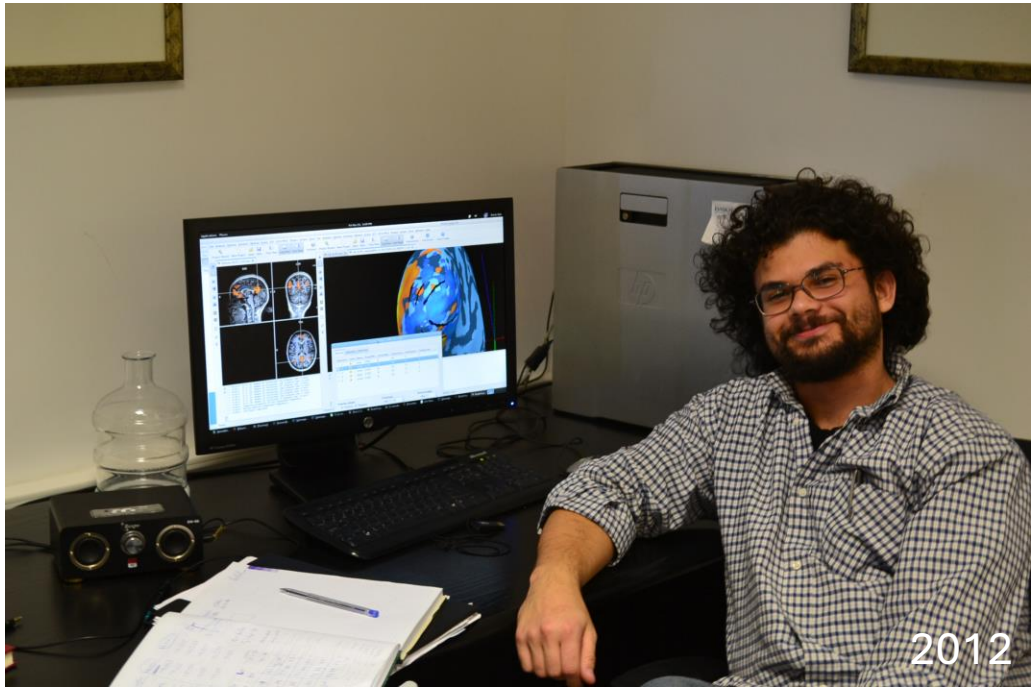




Skopelos
Skiros
Manisa
Izmir
Selçuk
Ephesus
Kuşadasi
Samos
Syros
Delos
Naxos
Paros
Sifnos
Antiparos
Sikinos
Ios
Amorgos
Astypalea
Tilos
Nissiros
Halki
Milos
Folegandros
Santorini
Anafi
Sea of
Karpathos
Kassos
Hania
Rethymnon
Heraklion
Sitia
CRETE
Agios Nikolaos
Phaestos
Gavdopoula
Gavdos

Background

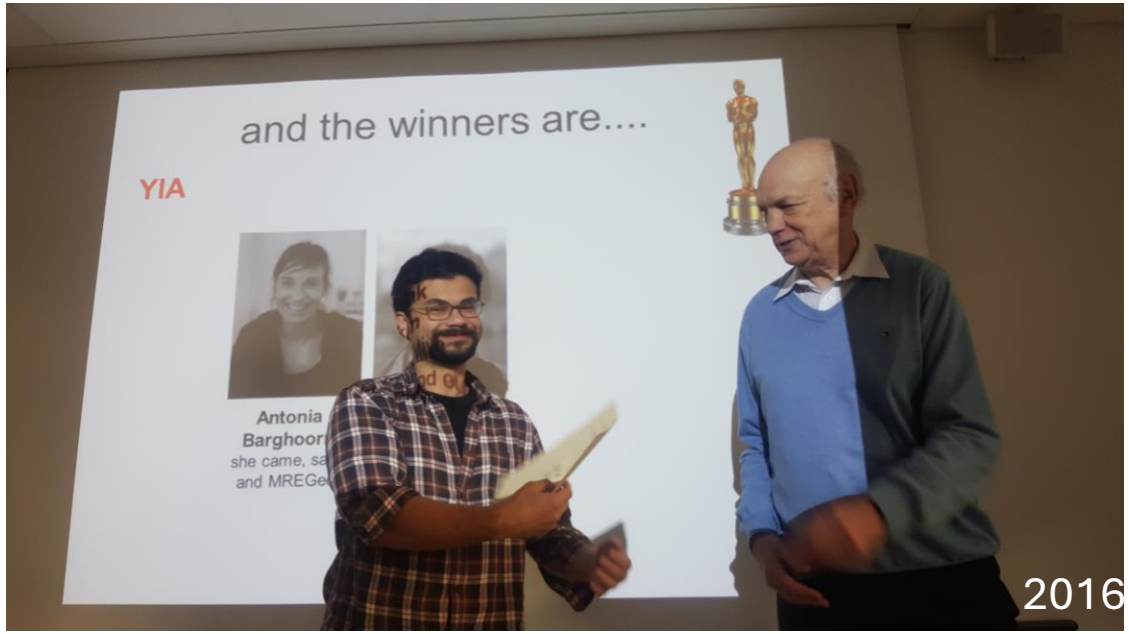




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Background





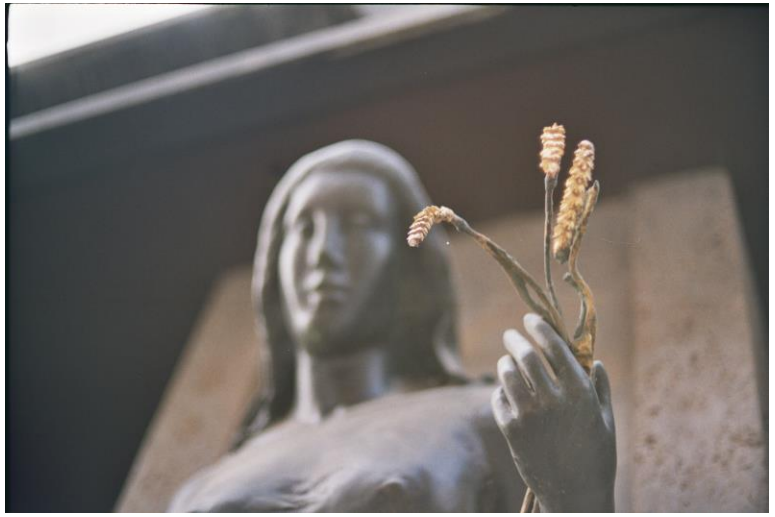
Background











Background









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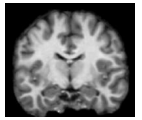
Background



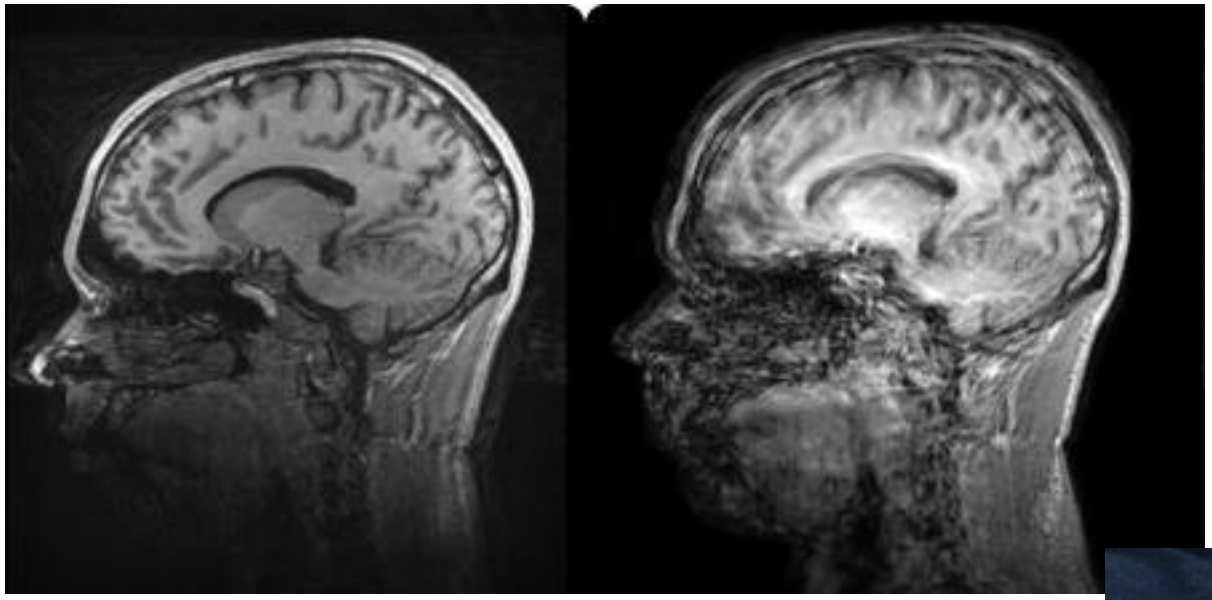


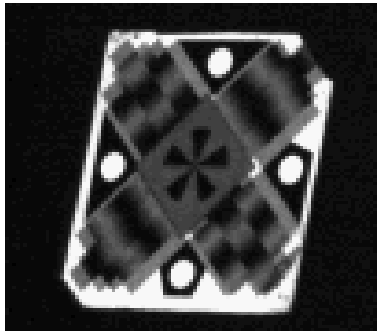
Year	2021	2022	2023	2024	2025	AVG
Distance	4.499,7 km	2.943,1 km	9.927,9 km	8.227,7 km	6.019,9 km	6323km
Vertical	39.497 m	38.320 m	106.702 m	81.518 m	69.234 m	67034m
Time	223h 53min	168h 24min	531h 23min	430h 45min	373h 12min	345h

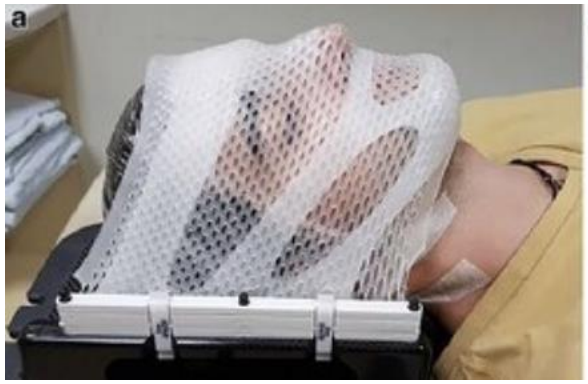
Motion in MRI



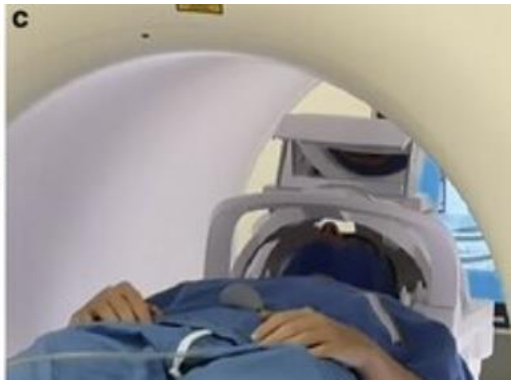








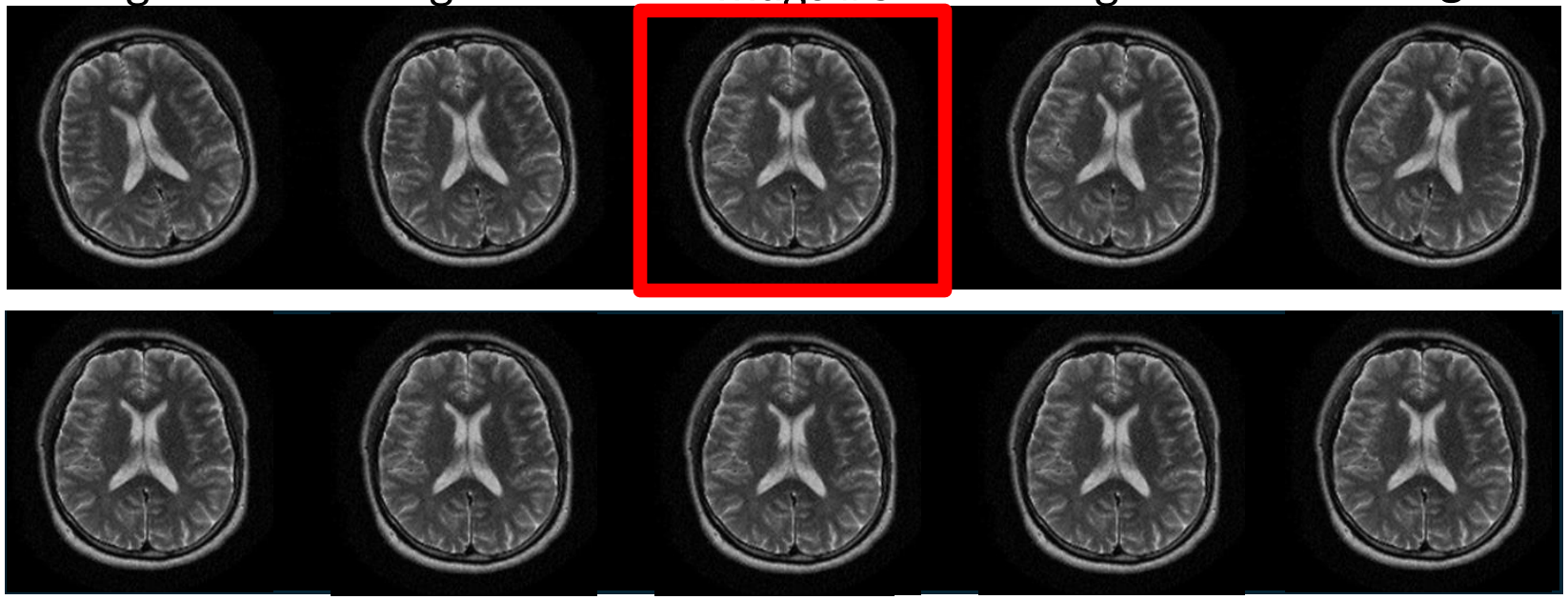


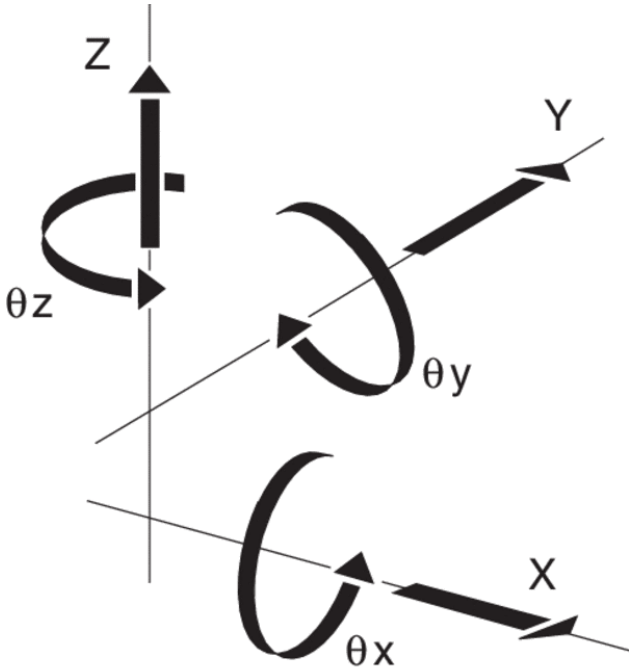




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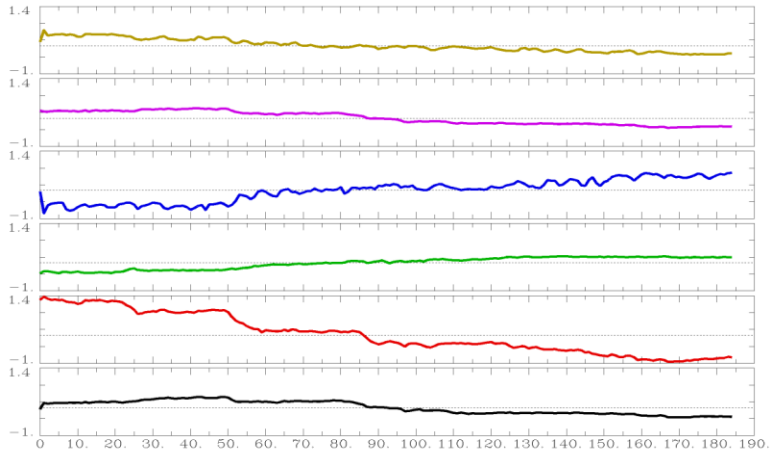
Rigid body motion correction



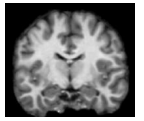


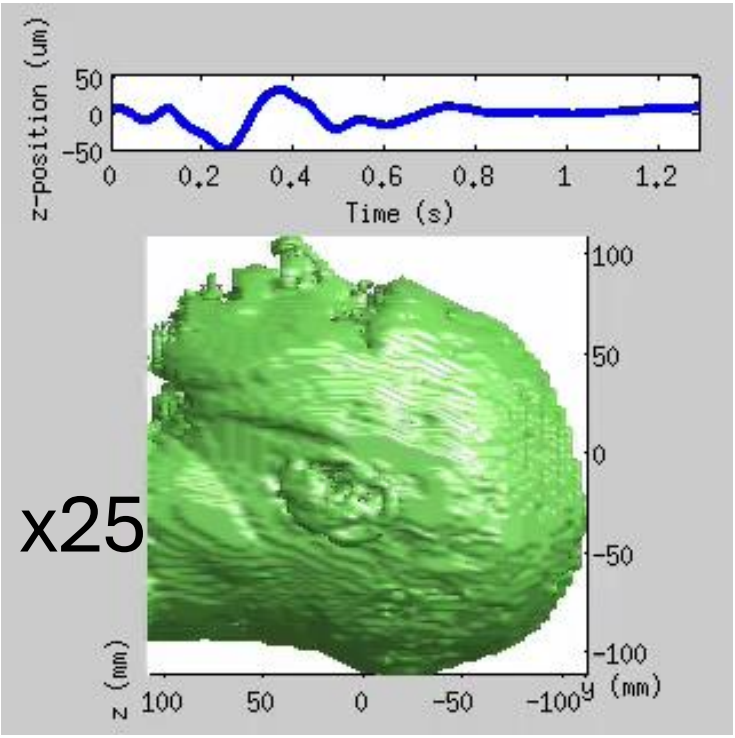



```
3dvolreg -verbose \ -zpad 1 \ -base
pb01.sub01.r01.tshift+orig'[0]' \ -1Dfile
dfile.sub01.r01.1D \ -prefix
pb02.sub01.r01.volreg \ -cubic \
pb01.sub01.r01.tshift+orig
```



Sources of Motion



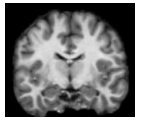


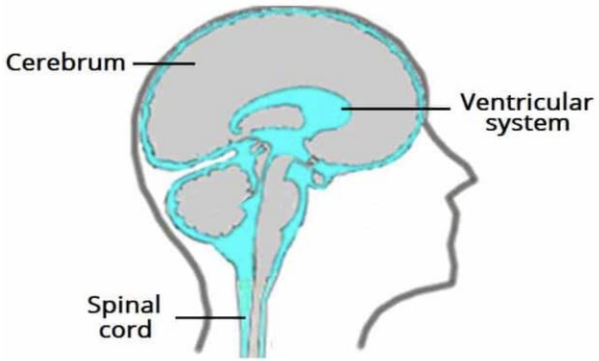


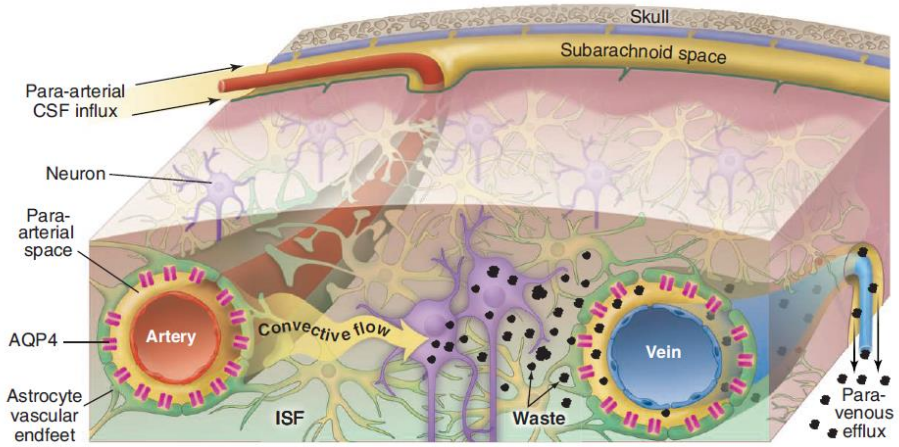
Craniotomy @ Northwestern Medicine

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Sources of Motion







Go with the flow. Convective glymphatic fluxes of CSF and ISF propel the waste products of neuron metabolism into the paravenous space, from which they are directed into lymphatic vessels and ultimately return to the general circulation for clearance by the kidney and liver.

Lymphatic draining system of the brain tissue, so called glymphatic system

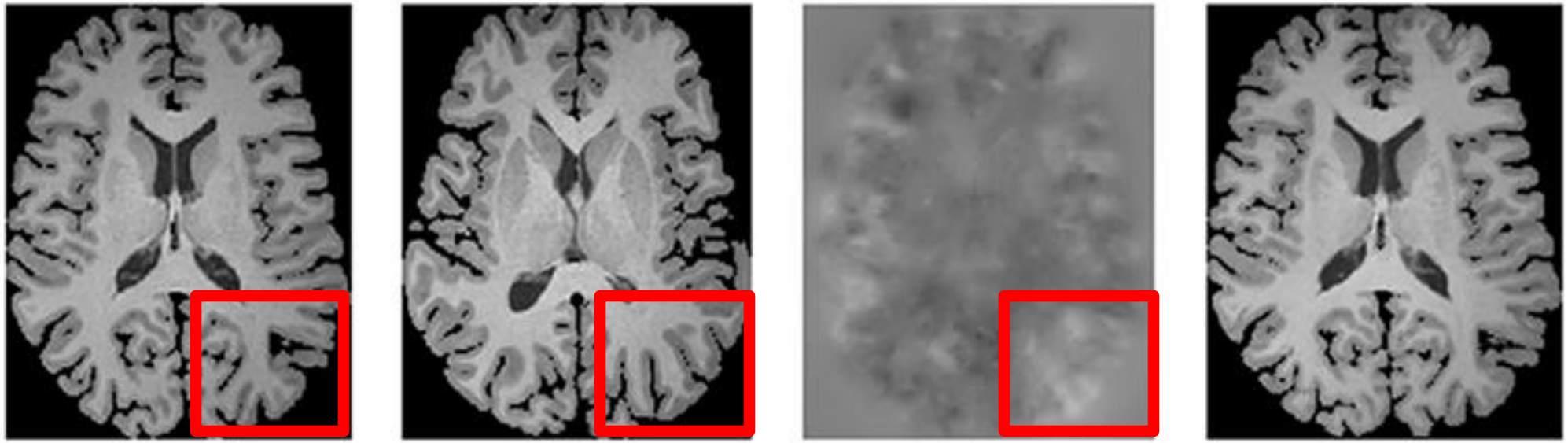
Arterial pulsations propulse water through aquaporin channels

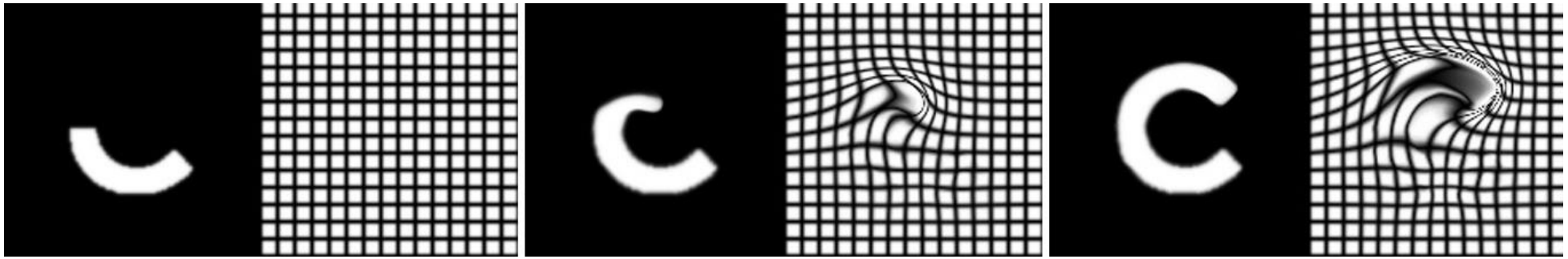
An intercellular “glymphatic” pathway clears cell waste from the brain and may reveal new targets for treating neurodegenerative diseases.

Garbage truck of the brain. Nedergaard M Science. 2013

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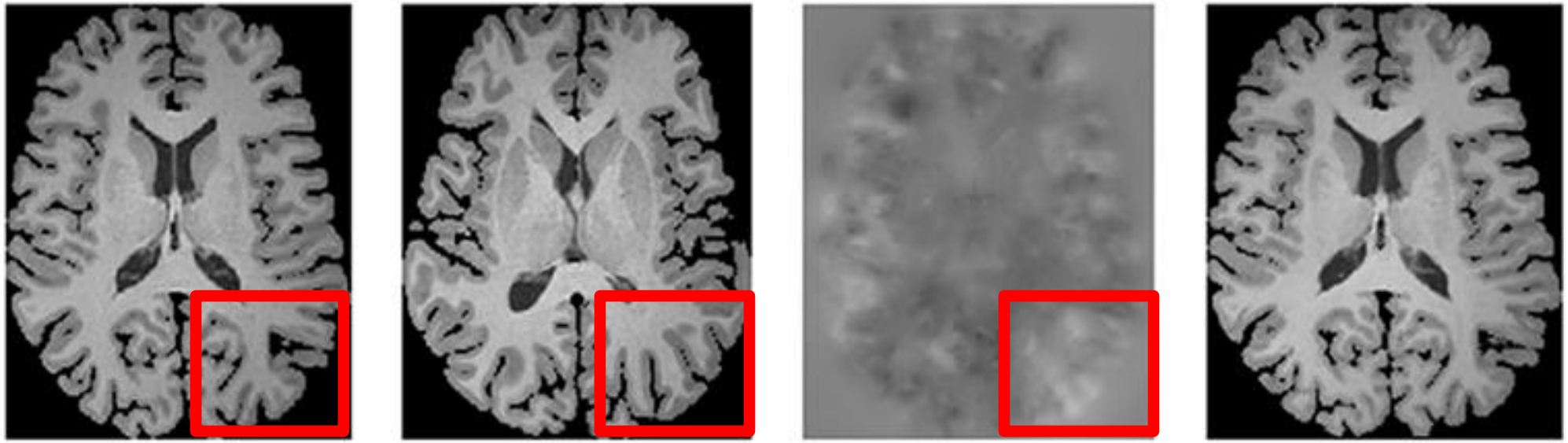
Non-linear registration





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Non-linear registration



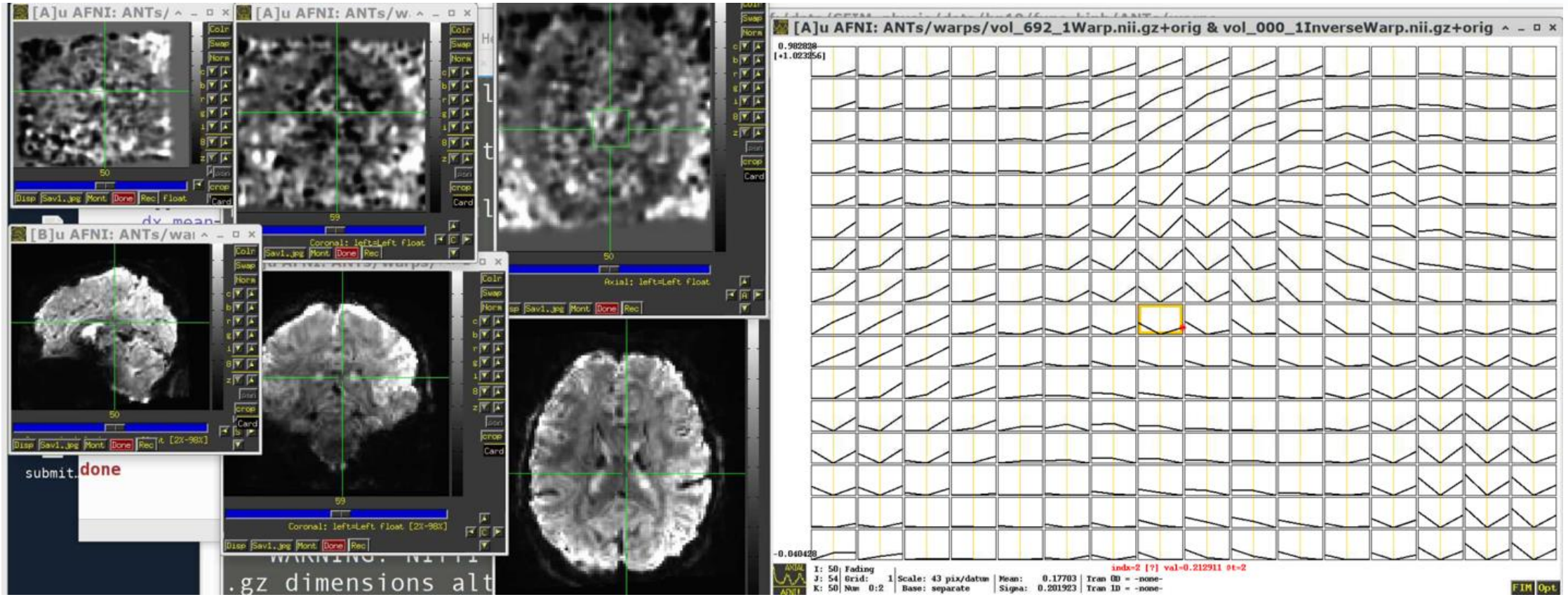
Fixed Moving Deformation Warped

```
antsRegistration \
  --dimensionality 3 \
  --float 1 \
  --collapse-output-transforms 1 \
  --output [ "${OUTDIR}/warps/vol_${v}_", "$OUTWARP", 0 ] \
  --interpolation Linear \
  --winsorize-image-intensities [0.01,0.99] \
  --use-histogram-matching 0 \
  --initial-moving-transform [ "$REF", "$MOV", 1 ] \
```

```
--transform SyN[0.03,2,0] \
--metric CC["$REF","$MOV",1,2] \
--convergence [10x5,1e-5,5] \
--shrink-factors 2x2 \
--smoothing-sigmas 1x1vox
```

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Image registration (T2* fMRI images)

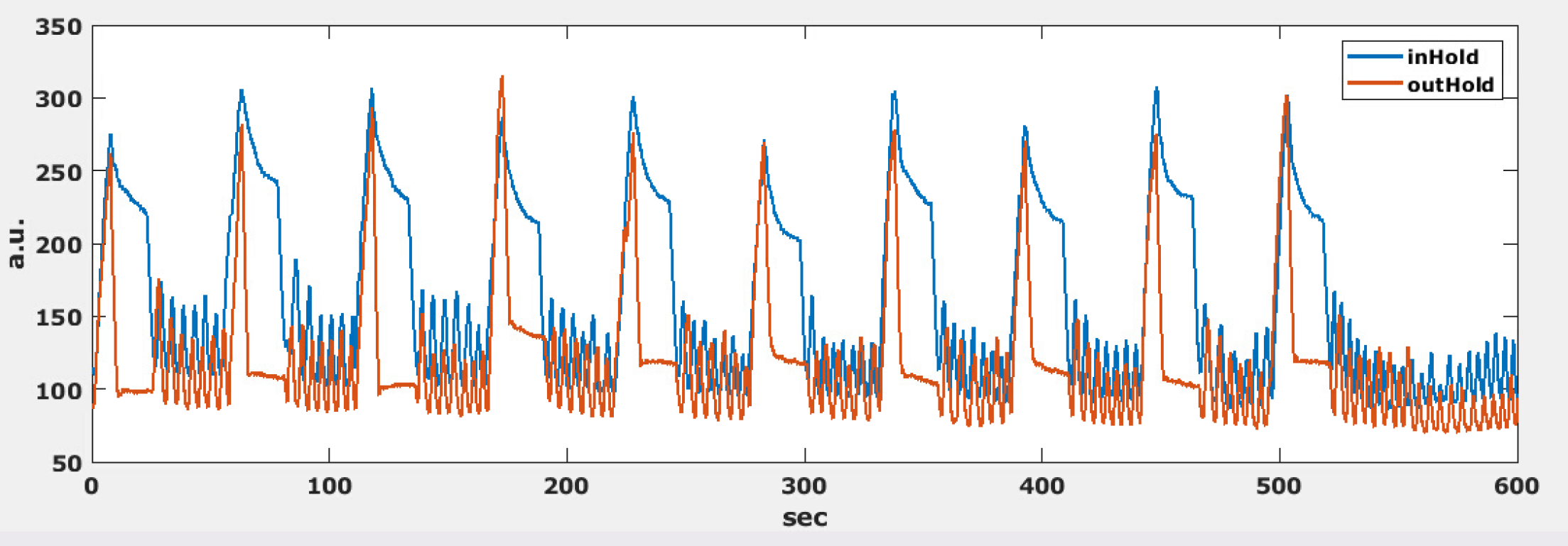


$$\text{Magnitude} = \sqrt{dx^2 + dy^2 + dz^2}$$

$$\theta = \arccos\left(\frac{dz}{\sqrt{dx^2 + dy^2 + dz^2}}\right)$$

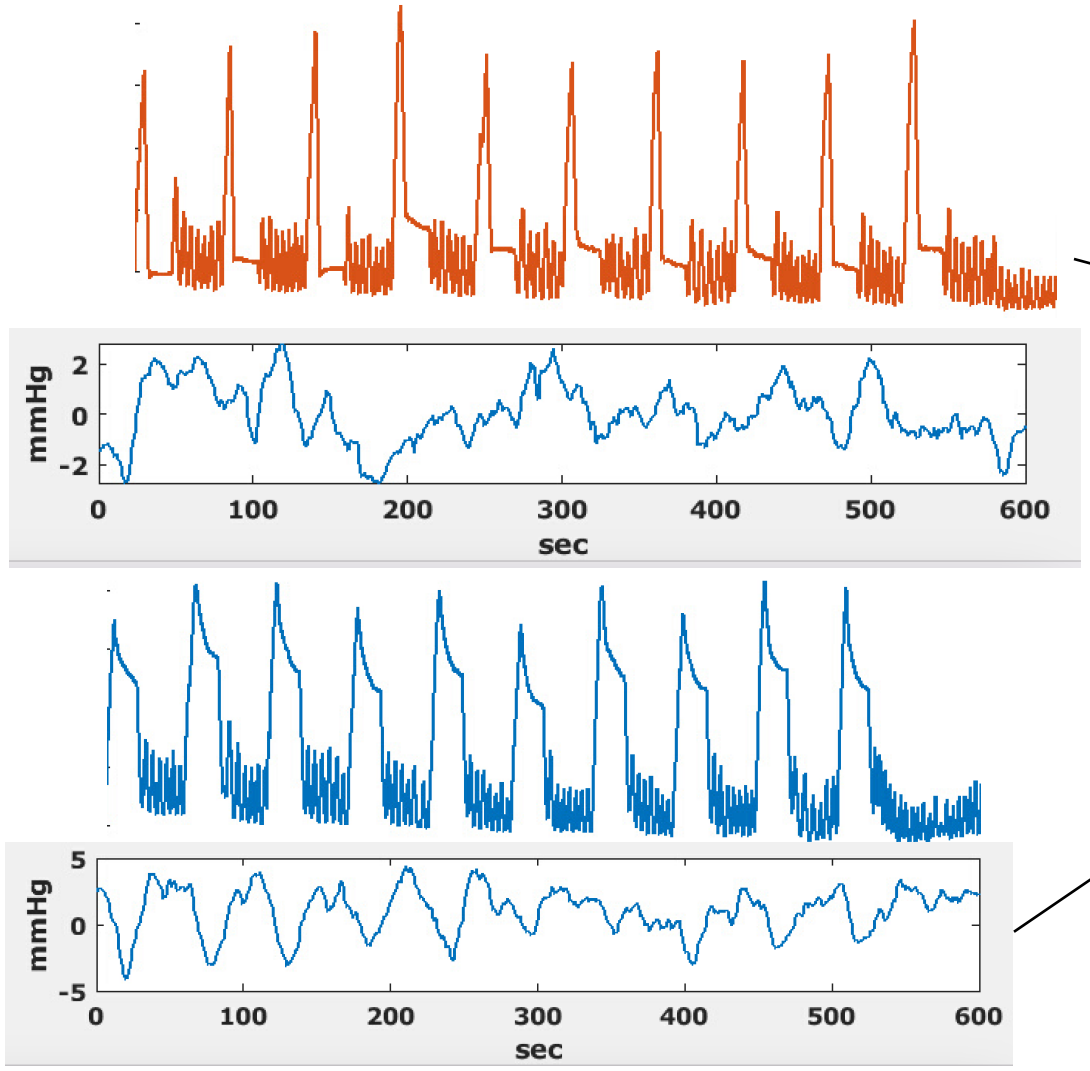
Breath Holding tasks

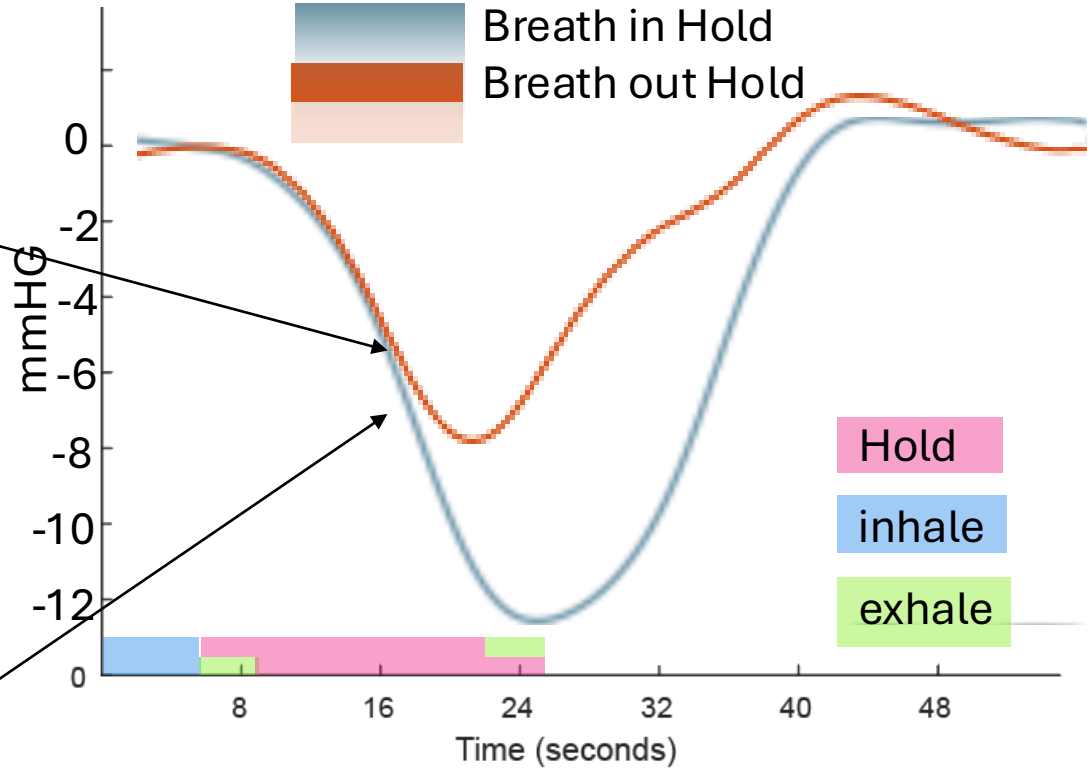
- Respiration volume changes during two task scans



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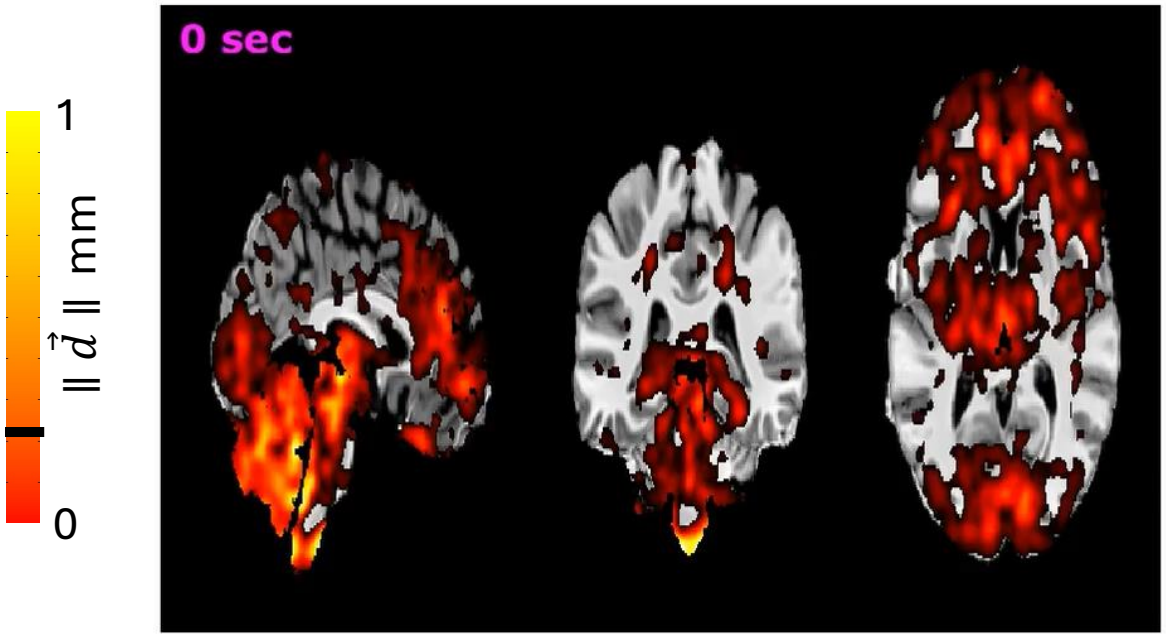
Trial Related Averaging MAP



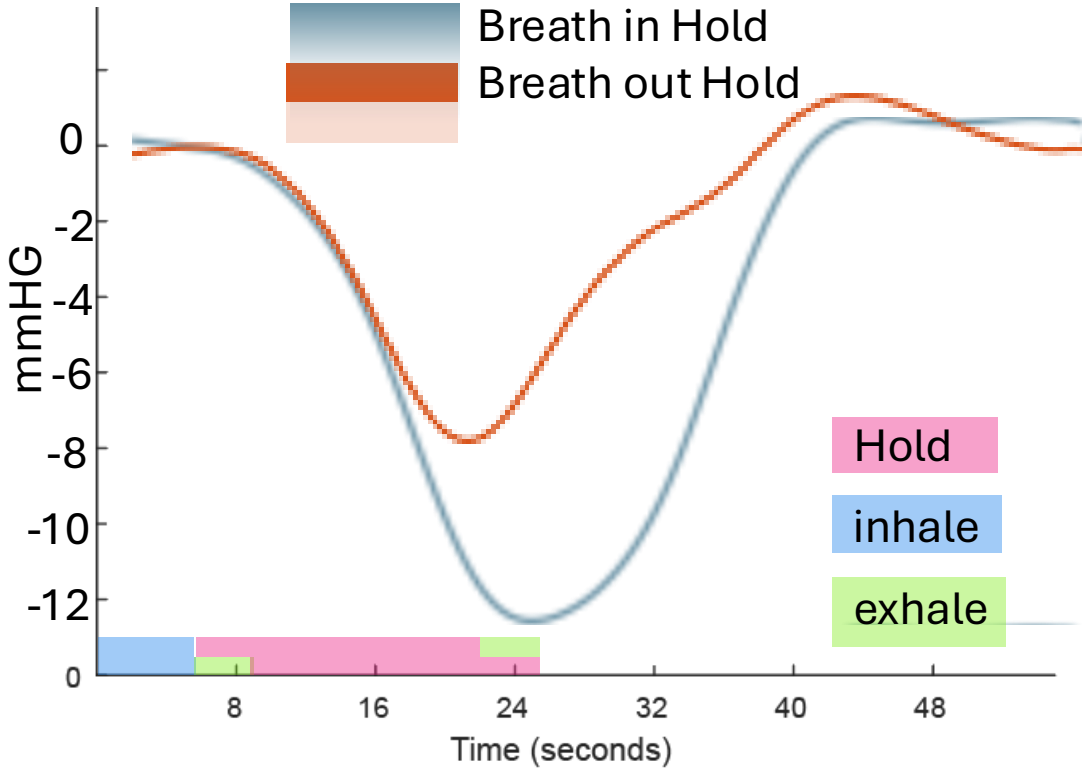


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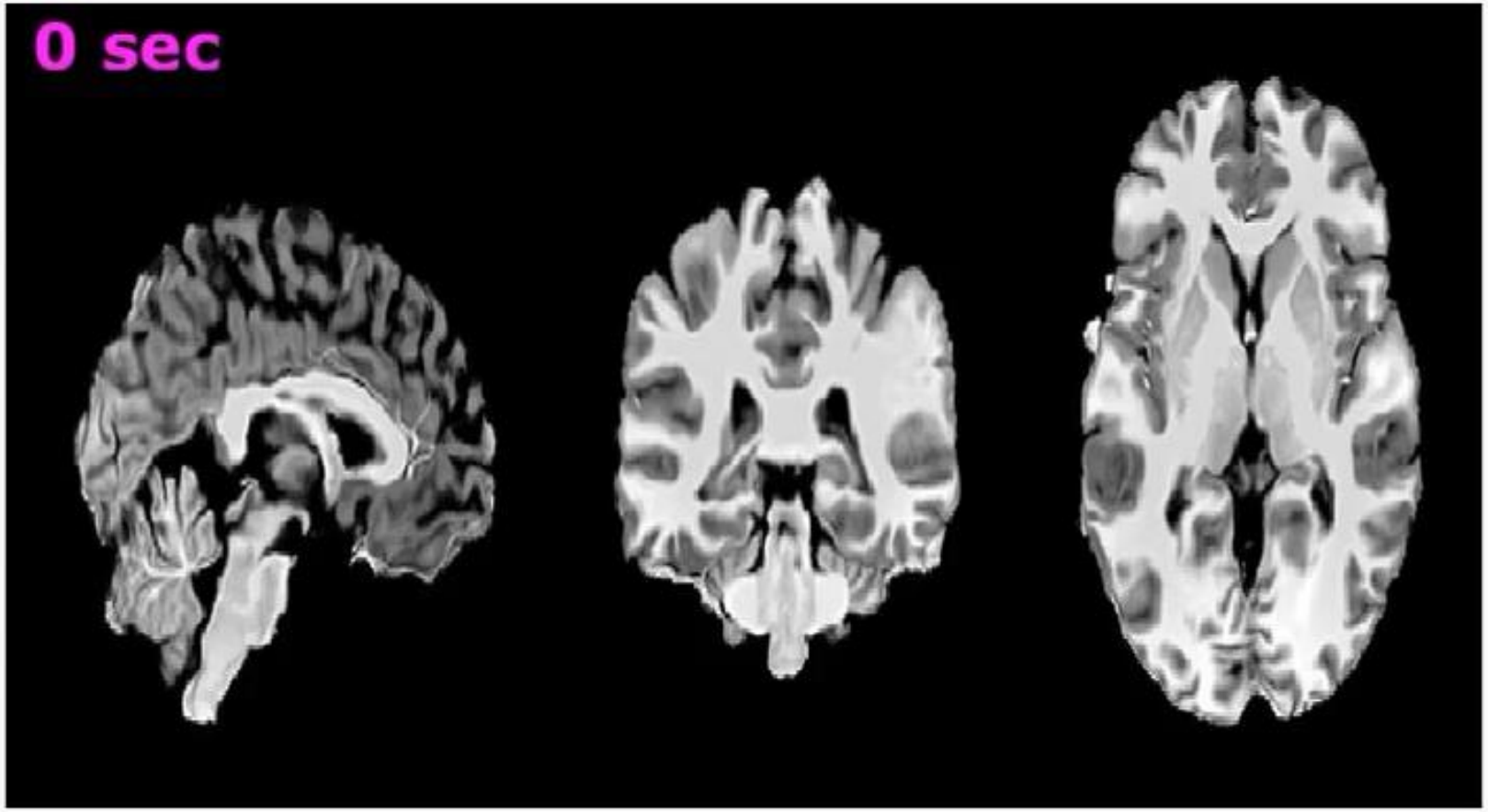
Task related displacement at physiological compartments



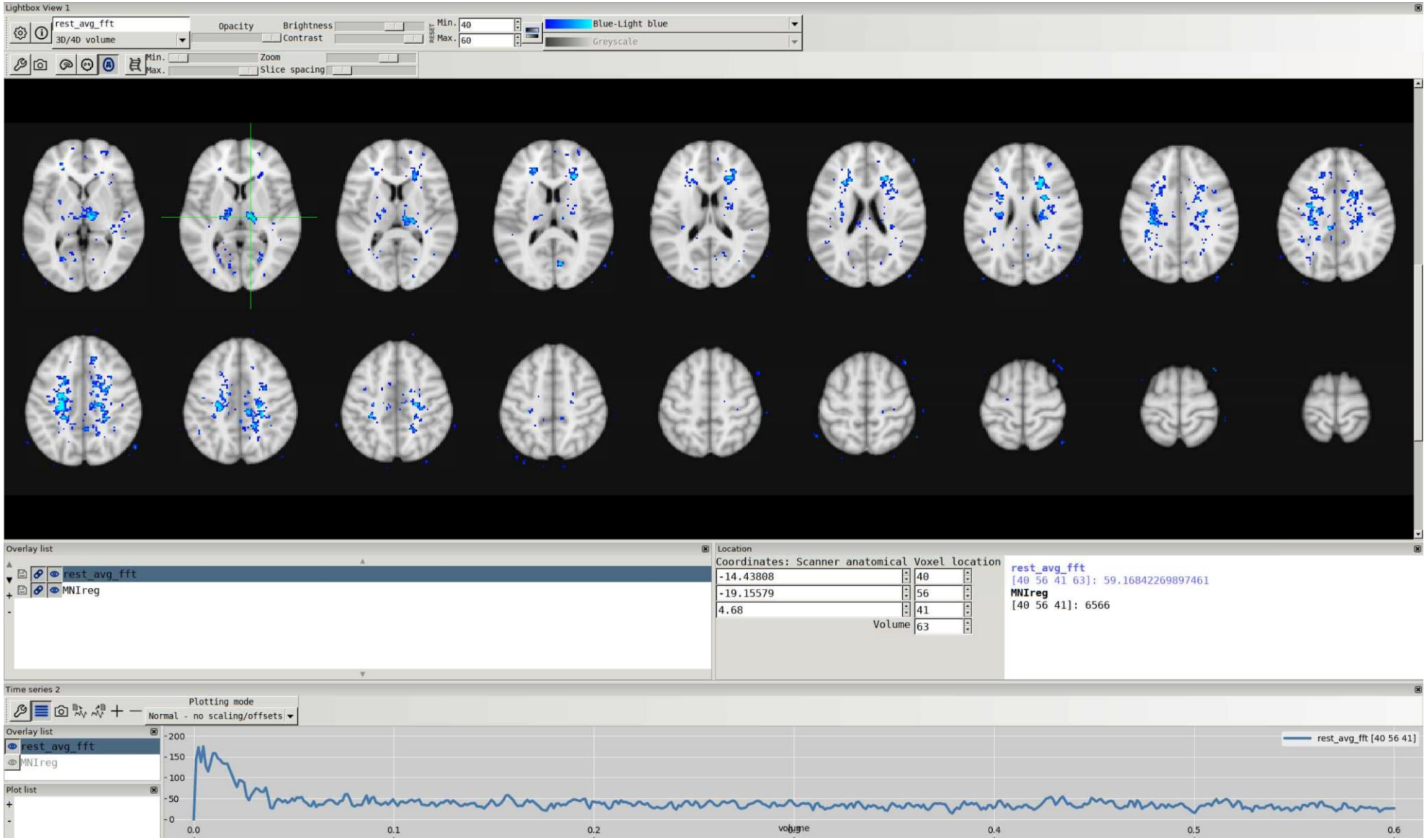
$$\|\vec{d}\| = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$



Task related displacement applied on T1 Anatomical x50



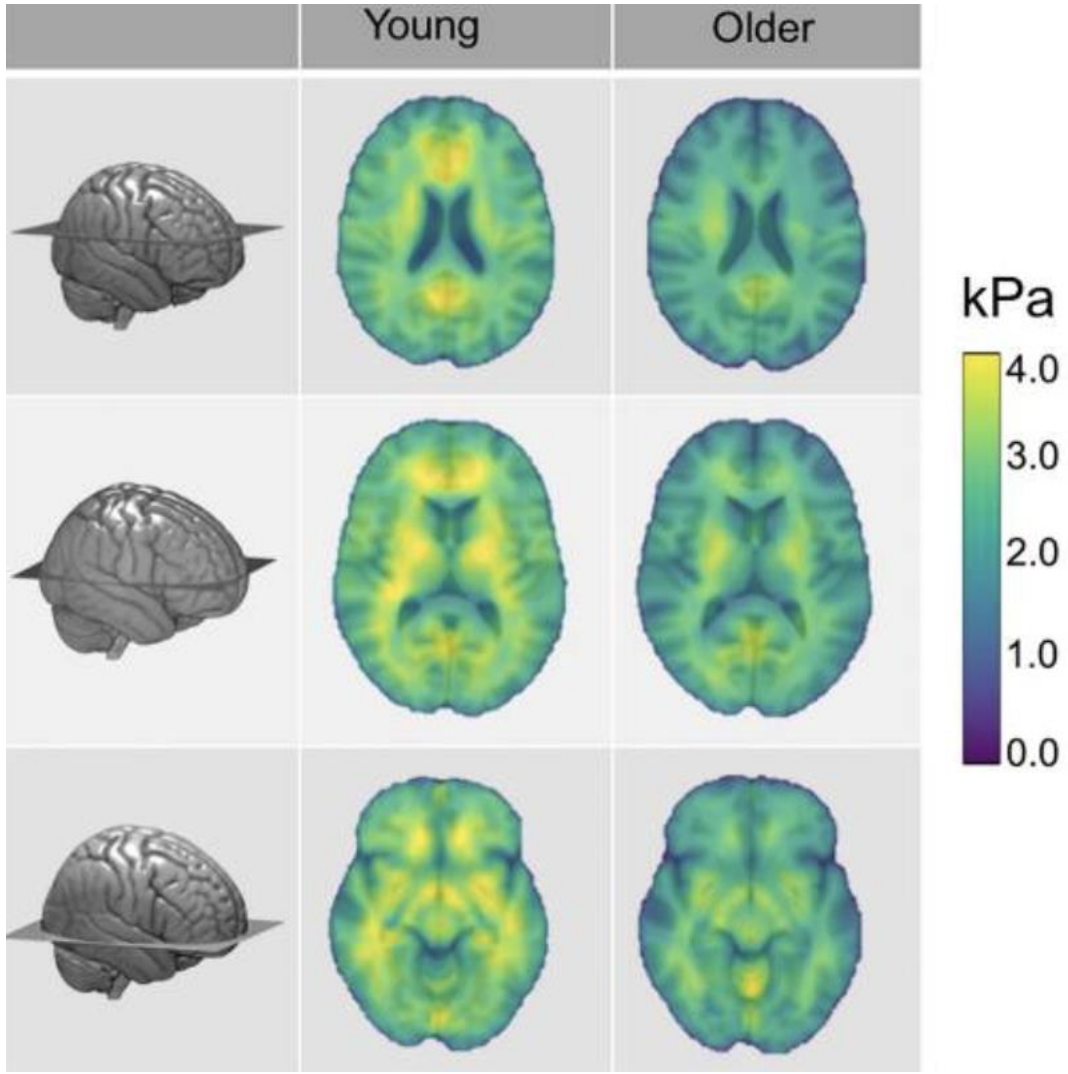
Resting state – Warp field displacements



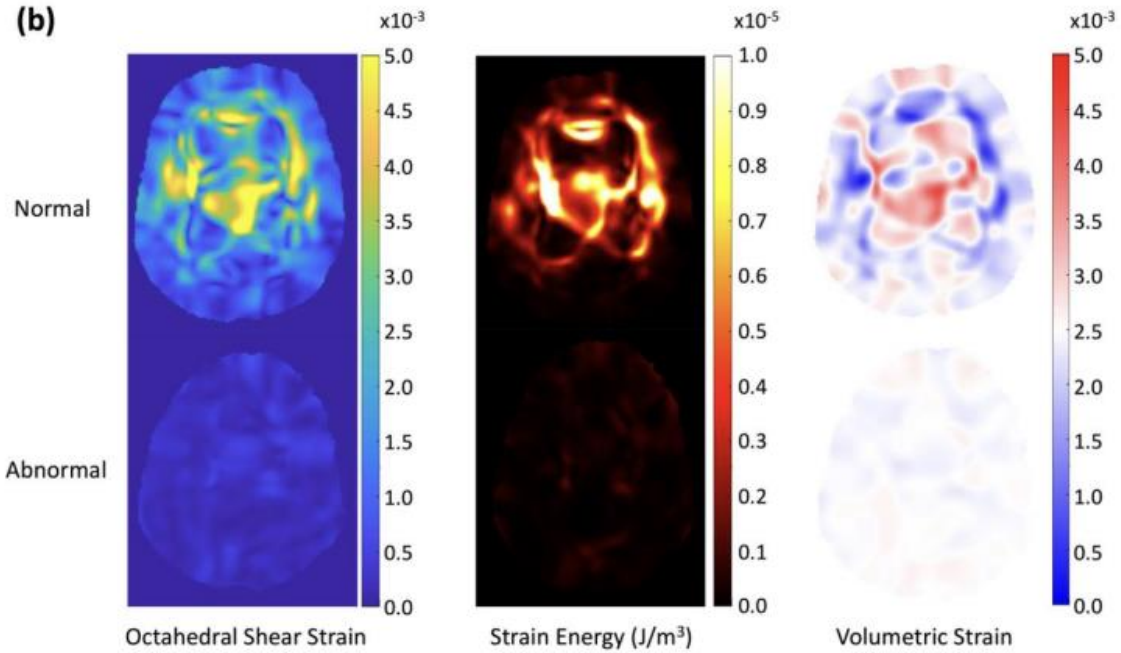
Breathing– Warp field displacements



Brain Stiffness, MR-Elastography



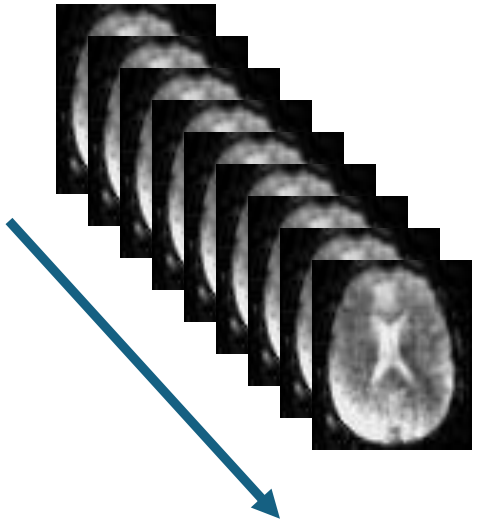
bSSFP -Insights from 3D q-aMRI

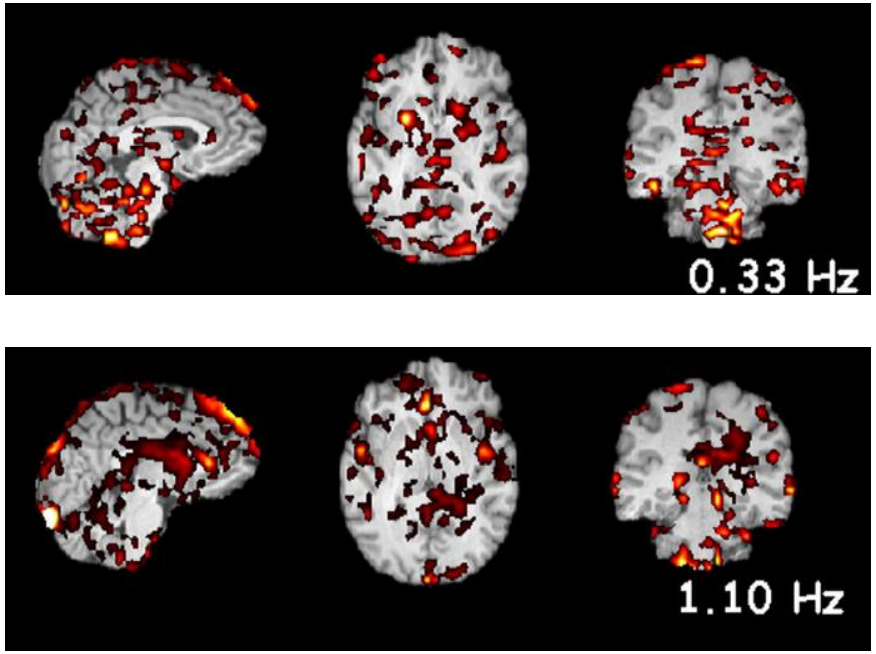


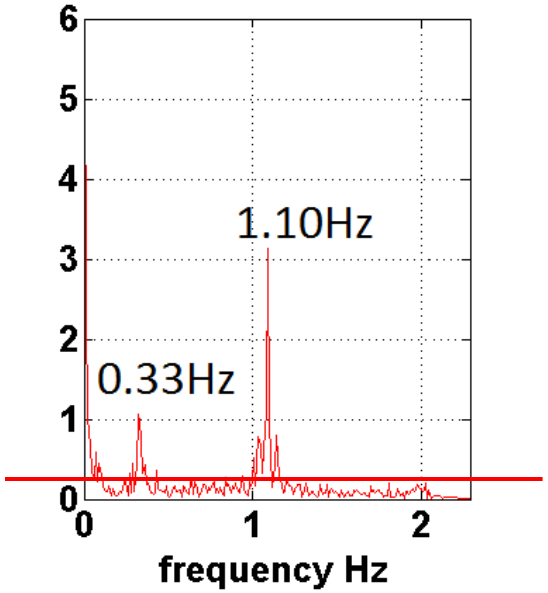
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Ultra fast fMRI TR=100ms at Resting state

- Non-linear warping every 100ms





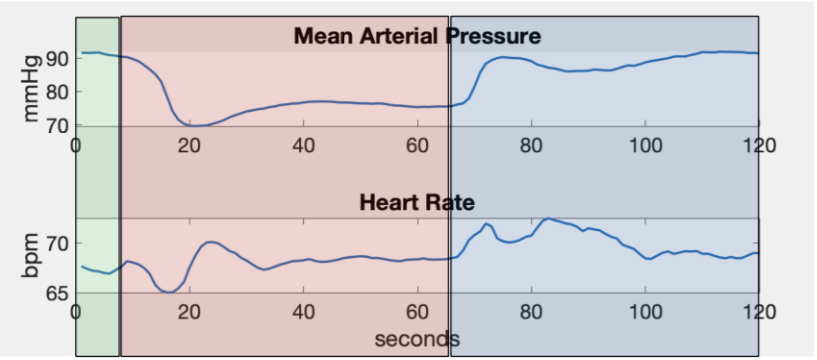


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Combining External measures

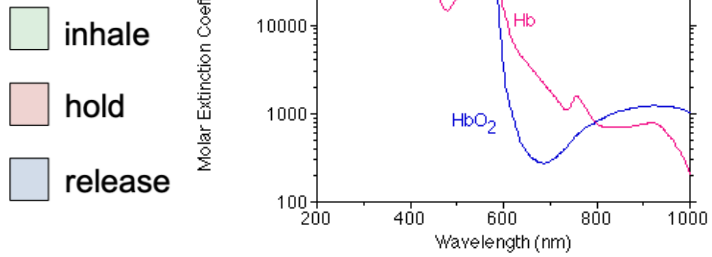
Blood Pressure Monitor (Caretaker)



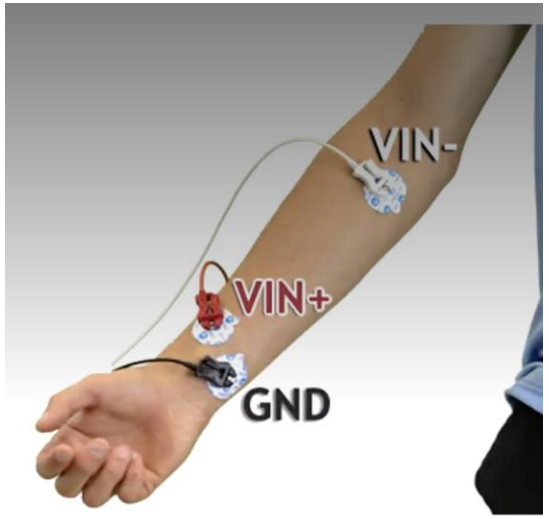


NIRS finger probe (Artinis)

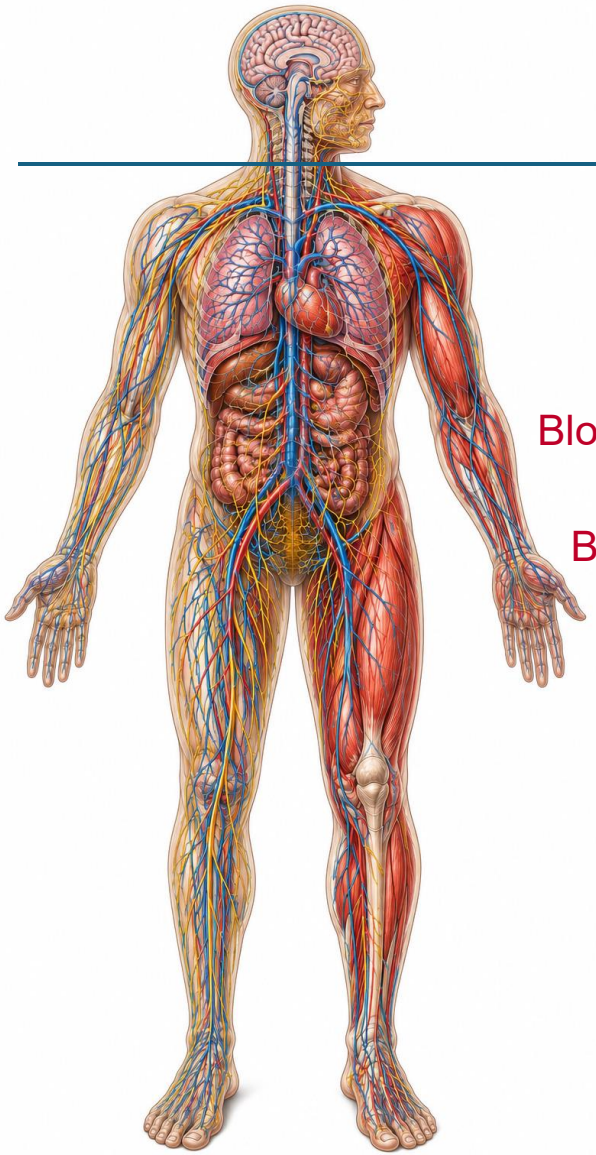




EMG monitor



Peripheral Measures



Blood Pressure

Blood Volume

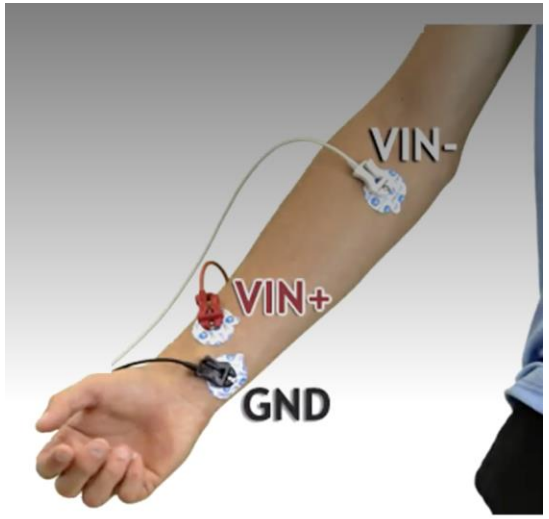
Muscle tone

Vascular tone

NIRS finger probe (Artinis)



EMG monitor

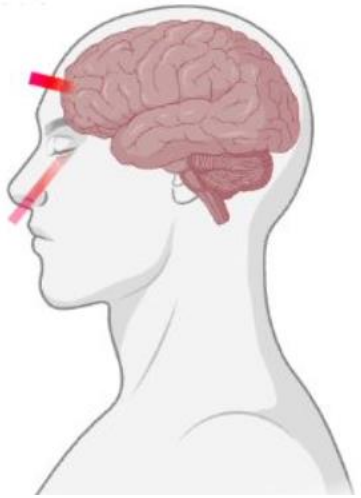


Blood Pressure Monitor (Caretaker)

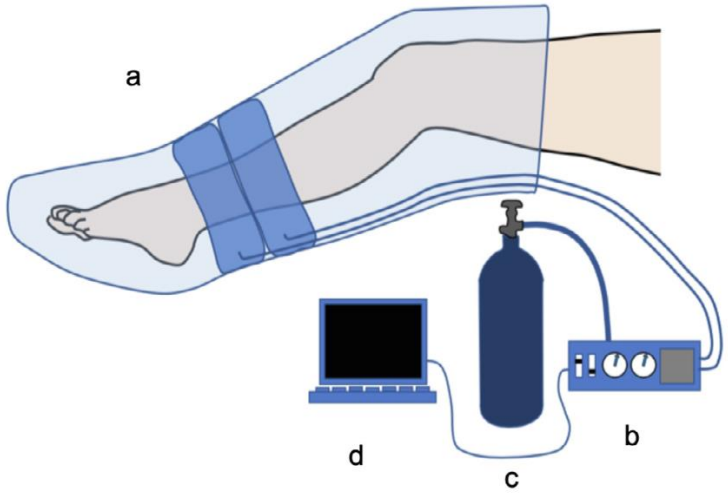


External stimulation 93M protocol

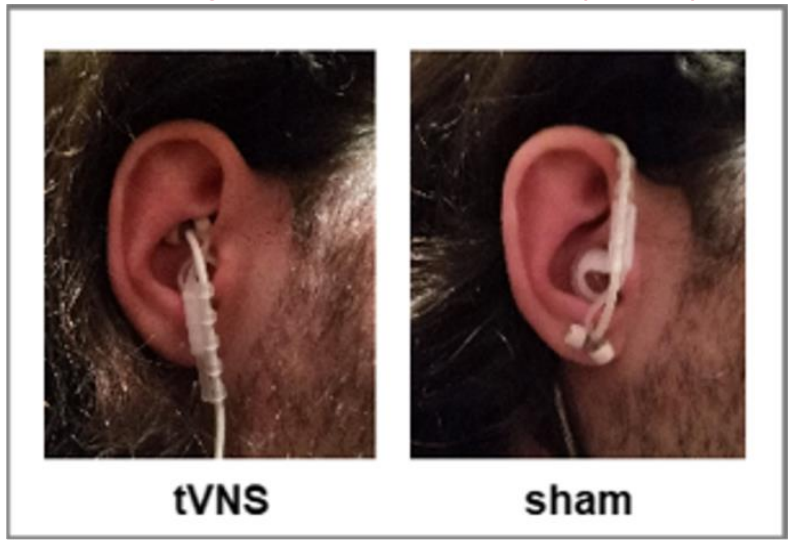
Light Stimulation



Pressure Cuff



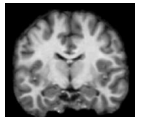
Vagal nerve stimulator(tVNS)



Obst et al., 2021



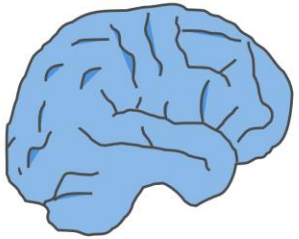
Thanks to colleagues…











Any questions ?



"OK, Mrs. Dunn. We'll slide you in there, scan your brain, and see if we can find out why you've been having these spells of claustrophobia."









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